

Customer Segmentation for a Retail Business That Has Recently Adapted An Online Store Utilizing the RFM Analysis and K-Means Clustering

Machine Learning Capstone Project

Dataset: UCI Online Retail Dataset

Student: Roberto G. Sanchez

Course: Post Graduate Diploma in Artificial Intelligence and Machine Learning

Instructor: Ritesh Chandra

Date: TBD

Executive Summary

This project aims to apply machine learning methodologies of customer segmentation to utilize transaction data from an online retailer. The objective of this project was to be able to identify different groups of customers that have similar purchasing behaviors and provide insightful business recommendations that can support customer retention and marketing strategies.

The data used for this study is the UCI Online Retail data set that contains 541,909 transaction records that covered December 2010 to December 2011. The features that were created include the Recency, Frequency and Monetary (RFM) for customer level analysis and were used as inputs for the K-means clustering. The findings of the analysis were highly skewed, logarithmic transformation and multiple steps to standardize the data were applied to the code before modelling.

The final K-Means model was able to identify two customer segments. The first cluster were the low engagement customers that consisted of 61.6% of the total data and contributed approx. 14.9% of the total monetary value. On the other hand, the high value segment consisted of 38.4% of the customers but contributed 85.1% of total customer monetary value. During the analysis, the model was able to achieve a Silhouette score of 0.4328 and a Davies-Bouldin Index of 0.8925.

The results from the analysis would suggest that the retailer should prioritize their high value customers through different loyalty and personalized customer incentive initiatives, while also aiming to re-engage with their lower engagement customers through different campaigns and strategies to convince these customers to purchase products again.

Business Problem

The business has recently adapted online retail and now has gotten a large amount of transaction data from their online platform. The problem is that they do not have any data-driven methods for identifying their customers' purchasing behaviors in order to make decisions on which customers they should prioritize for retention and targeted marketing.

Data Science Problem

This project is utilizing the unsupervised machine learning technique, specifically a customer clustering/segmentation task. K-means clustering was used in the project in order to group customers based on their similarities in their purchasing behavior.

Project Objective

The objectives of this project are to:

- Clean and prepare the transaction dataset.
- Develop customer level RFM.
- Use the K-Means clustering method to identify the appropriate customer segments.
- Determine the quality and stability of the resulting clusters.
- From the identified clusters and data, interpret each of the characteristics.
- Create impactful, insightful and meaningful business recommendations based on the data gathered.

KPI to Achieve:

- Silhouette Score - Higher is better
- Davies-Bouldin Index - Lower is better
- Inertia - Used with the elbow method

These will help us achieve better customer targeting, identification of high value customers, and improved customer retention.

Dataset Source

The project uses the data from the Online Retail Dataset from the UCI Machine Learning Repository. The data set is able to provide crucial information like transaction level records from an online retail business and is most suitable for analyzing the purchasing behavior of customers.

As mentioned previously, the dataset contains 541,909 transaction records and 8 different variables. The dataset also covers the period from December 1, 2010 to December 9, 2011.

RFM Dataset

After preprocessing, data cleaning and customer level clustering, the final RFM dataset constrained 4,338 customers.

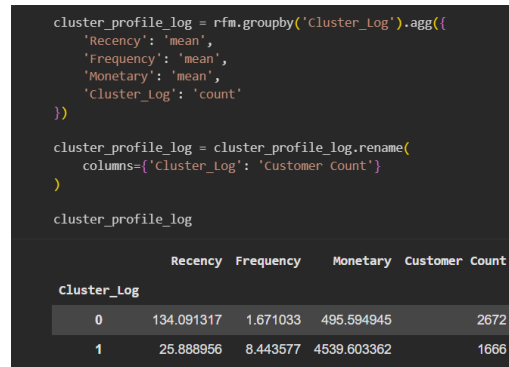


Figure 1. RFM results of the UCI Online Retail Dataset

The three features used for the clustering were:

- Recency: days since the customers most recent purchase.
- Frequency: the number of unique invoices/purchases made by the customer.
- Monetary: the total customer spending calculated from quantity multiplied by the unit price.

Data Preprocessing

There were multiple preprocessing steps that were performed before the clustering was done. Firstly, the transactions that had negative quantities were removed because they represented returns or canceled transactions rather than completed purchases, so this was something more desirable to remove from the dataset as it supplied no crucial information and could have just introduced noise to the data. Records that showed zero or negative unit prices were also removed from the dataset because they did not represent valid positive sales and served no purpose in being included in the dataset.

Transactions that had no valid customer ID were also excluded because they could not be reliably associated with an individual customer and could also contribute to a discrepancy in the data. Products with missing product descriptions proved not to be critical to the RFM analysis because the product description was not needed in the clustering features.

Handling Skewed Data

The initial RFM analysis showed a substantial value in right skewness, especially for the frequency and monetary values. Monetary values ranged from £3.75 to £280,206.02 while the frequency ranged from 1 to 209 in purchases.

In order to reduce the influence of these extreme values without compromising or deleting potentially legitimate high value customers, a logarithmic transformation was applied to the variables in

the RFM analysis. After the transformation, standardization was required to have comparable influence during the K-Means clustering.

```
print("Monetary:")
print(rfm['Monetary'].describe())

print("\nFrequency:")
print(rfm['Frequency'].describe())

print("\nRecency:")
print(rfm['Recency'].describe())
```

```
*** Monetary:
count      4338.000000
mean       2048.688081
std        8995.230220
min         3.750000
25%        306.482500
50%        668.570000
75%       1660.597500
max       280206.020000
Name: Monetary, dtype: float64

Frequency:
count      4338.000000
mean         4.272015
std         7.697998
min         1.000000
25%         1.000000
50%         2.000000
75%         5.000000
max        209.000000
Name: Frequency, dtype: float64

Recency:
count      4338.000000
mean        92.536422
std        100.014169
min         1.000000
25%        18.000000
50%        51.000000
75%       142.000000
max       374.000000
Name: Recency, dtype: float64
```

Figure 2. Descriptive Statistics showing the skewness and extreme values in customer RFM features

Final Model Evaluation

After standardizing and transforming the RFM features of the two clusters of the final K-Means model, it was able to achieve a Silhouette score of 0.4328 and a Davies-Bouldin index of 0.825. These results are indicating that there is a moderate separation between the customer groups.

In order to visualize the data, the PCA was also applied. The first component was able to explain 75.08% of the variance, while the second explained the 18.79%. When combined, the two values retained 93.87% of the total variance and was able to provide a two dimensional representation of the customer data which is useful to the study.

```
print("Final model:")
print("Number of clusters:", best_k_log)
print("Silhouette Score:", round(final_silhouette_log, 4))
print("Davies-Bouldin Index:", round(final_db_log, 4))
print("PCA variance retained:", round(pca_final.explained_variance_ratio_.sum() * 100, 2), "%")
```

```
Final model:
Number of clusters: 2
Silhouette Score: 0.4328
Davies-Bouldin Index: 0.8925
PCA variance retained: 93.87 %
```

Figure 4. Final K-Means evaluation metrics.

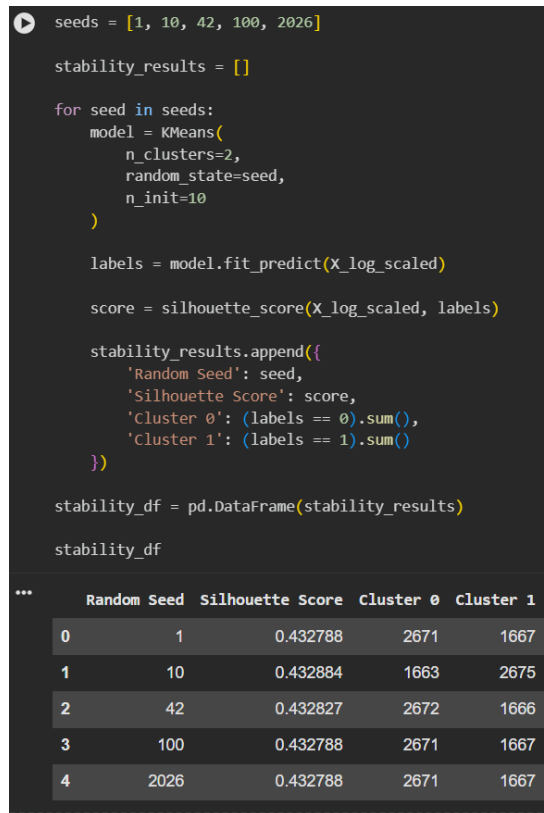


Figure 5. K-Means stability across different random seeds.

Customer Segmentation Results

The final model identified two distinct customer segments.

Metric	Cluster 0	Cluster 1
Customer Count	2,672	1,666
Customer Share	61.60%	38.40%
Average Recency	134.1 days	25.9 days
Median Recency	96 days	16 days
Average Frequency	1.67	8.44
Median Frequency	1	6
Average Monetary	£495.59	£4,539.60
Median Monetary	£363.08	£2,061.08
Monetary Share	14.90%	85.10%

Cluster 0: The Low Engagement Customers

Cluster 0 consists of 2,672 customers and represents 61.6% of the customer population. These customers have a higher average recency of 131.1 days, a low average purchase frequency of 1.67 and an average monetary value of £495.59. This group represents the larger population of customers however, they only contribute to 14.9% of the total customer monetary value. The segment therefore represents an opportunity for retention strategies and the business should explore opportunities to convert these customers into high engagement customers.

Cluster 1: High Value Engaged Customers

Cluster 1 contains 1,666 customers and represents 38.4% of the customer population. These are the customers that have a much lower average recency of 25.9 days, they purchase more frequently with an average of 8.44, and they have an even greater average monetary value of £4,539.60 compared to the low engagement customers. Although they only represent 38.4% of the customers, this cluster contributes approximately 85.1% of the total monetary value. This means that this cluster is very significant for the business and major effort should be put into making sure that they are retained with loyalty incentives and other initiatives to boost customer loyalty.

Bias and Fairness Analysis

It was challenging to determine bias and fairness due to the fact that the dataset did not contain sensitive demographic characteristics such as age, gender, race or income of the customers. The chosen characteristic in order to determine the bias and fairness was demographic parity and equalized odds could not be directly calculated.

Geographic representation was analyzed using the country variable from the dataset and proved to be a potential source of bias. Customer groups from the United Kingdom, France, and Germany were much larger variables compared to the rest in both cluster 0 and 1, as opposed to being exclusive to a single cluster.

Country	Cluster 0	Cluster 1
UK	45.83%	54.17%
Switzerland	51.72%	48.28%
Spain	52.13%	47.87%
Germany	64.29%	35.71%
France	65.00%	35.00%
Belgium	61.94%	38.06%

More on that, the countries with smaller values were mostly if not all assigned to one cluster. These results should not be interpreted as evidence though, as mentioned they had very small counts which should not serve as concrete evidence of systematic geographic discrimination.

With all of that being said, this classification should not be used to support discriminatory pricing, exclusion or denial of service but should be used to focus more on customer engagement and personalizing the marketing strategies of the company.

Business Recommendations

Based on the results from the research, different strategies or approaches on each cluster group.

For the high value customers that represent the smaller amount of the population but generate the majority of the monetary value. This should be the priority for the business and mainly focus their resources into retaining this customer group. I would recommend that the business introduce loyalty or VIP programs to provide incentives to their customers who are consistently purchasing items from their business. I also recommend providing personalized product recommendations on their online platform, as well as offering early access to new products and promotions.

In the case of the low engagement customers that represent the larger amount of the population but they spend significantly less than the high value customers. This should not be seen as a disadvantage though, but as an opportunity to convert these customers into the high value segment. It is feasible to see this as an opportunity to convert these customers through targeted win back campaigns, provide personal discounts for returning customers, have email reminders on promotions and discounts to encourage repeat purchases.

As mentioned the organization should not apply the same strategy for all the customers in the different clusters. Resources should obviously be prioritized into the high value customers while applying a targeted campaign to encourage the lower engagement customers to have a more frequent purchasing behavior towards the business and its products.

Conclusion

This project was able to demonstrate the end-to-end machine learning lifecycle for a customer segmentation research using the dataset from UCI Online Retail. The data used was transaction level data that was properly cleaned and transformed to include RFM features before a logarithmic transformation and standardization was applied to the data.

The K-Means cluster was used across the multiple values of K, and the final model was identified to be a K=2 model and was selected based on the quality, stability and interpretability of the clustering. The model was able to identify the low engagement cluster (cluster 0) that consisted of 61.6% of the

customer population while the high value engagement cluster (cluster 1) consisted of 38.4% of the customer population.

The most notable business insight was that the high value segment makes up the majority of the monetary accounts (85.1%) despite consisting only a small amount of the total population of customers (38.4%). Frequency was also identified to be a strong feature that separated the two clusters, showing that one cluster was more frequent in their purchasing behavior compared to the other cluster that was not purchasing as frequently.

Overall the result demonstrates how an unsupervised machine learning model can transform simple and unorganized transaction data into actionable objectives in order to boost customer engagement. The resulting segments can also support in making important business decisions and provide justification for resource allocation into customer loyalty and incentive programs, allowing the business to know if it is worth the cost.